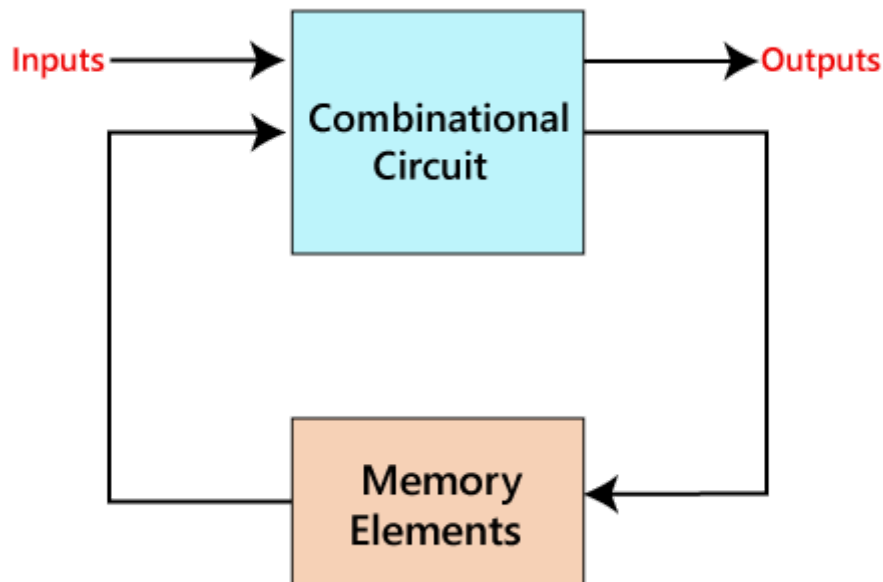


Sequential Circuits

In our previous sections, we learned about combinational circuit and their working. The combinational circuits have set of outputs, which depends only on the present combination of inputs. Below is the block diagram of the synchronous logic circuit.



The sequential circuit is a special type of circuit that has a series of inputs and outputs. The outputs of the sequential circuits depend on both the combination of present inputs and previous outputs. The previous output is treated as the present state. So, the sequential circuit contains the combinational circuit and its memory storage elements. A sequential circuit doesn't need to always contain a combinational circuit. So, the sequential circuit can contain only the memory element.

Difference between the combinational circuits and sequential circuits are given below:

	Combinational Circuits	Sequential Circuits
1)	The outputs of the combinational circuit depend only on the present inputs.	The outputs of the sequential circuits depend on both present inputs and present state (previous output).
2)	The feedback path is not present in the combinational circuit.	The feedback path is present in the sequential circuits.

3)	In combinational circuits, memory elements are not required.	In the sequential circuit, memory elements play an important role and require.
4)	The clock signal is not required for combinational circuits.	The clock signal is required for sequential circuits.
5)	The combinational circuit is simple to design.	It is not simple to design a sequential circuit.

Types of Sequential Circuits

Asynchronous sequential circuits

The clock signals are not used by the **Asynchronous sequential circuits**. The asynchronous circuit is operated through the pulses. So, the changes in the input can change the state of the circuit. The asynchronous circuits do not use clock pulses. The internal state is changed when the input variable is changed. The un-clocked flip-flops or time-delayed are the memory elements of asynchronous sequential circuits. The asynchronous sequential circuit is similar to the combinational circuits with feedback.

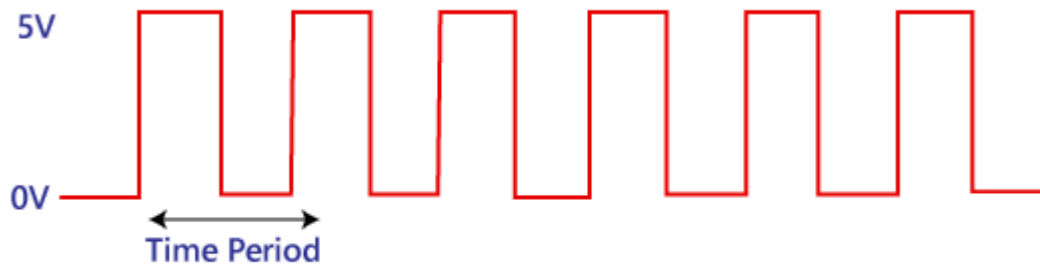
Synchronous sequential circuits

In synchronous sequential circuits, synchronization of the memory element's state is done by the clock signal. The output is stored in either flip-flops or latches(memory devices). The synchronization of the outputs is done with either only negative edges of the clock signal or only positive edges.

Clock Signal and Triggering

Clock signal

A clock signal is a periodic signal in which ON time and OFF time need not be the same. When ON time and OFF time of the clock signal are the same, a square wave is used to represent the clock signal. Below is a diagram which represents the clock signal:



A clock signal is considered as the square wave. Sometimes, the signal stays at logic, either high 5V or low 0V, to an equal amount of time. It repeats with a certain time period, which will be equal to twice the 'ON time' or 'OFF time'.

Types of Triggering

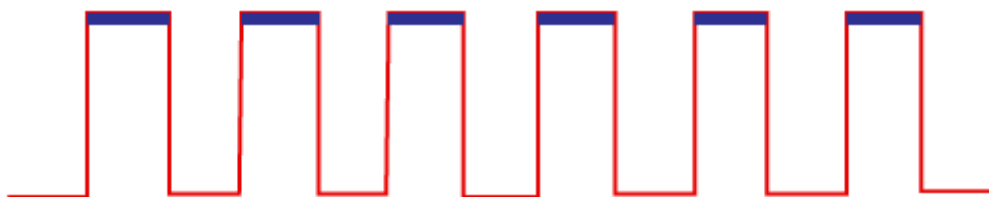
These are two types of triggering in sequential circuits:

Level triggering

The logic High and logic Low are the two levels in the clock signal. In level triggering, when the clock pulse is at a particular level, only then the circuit is activated. There are the following types of level triggering:

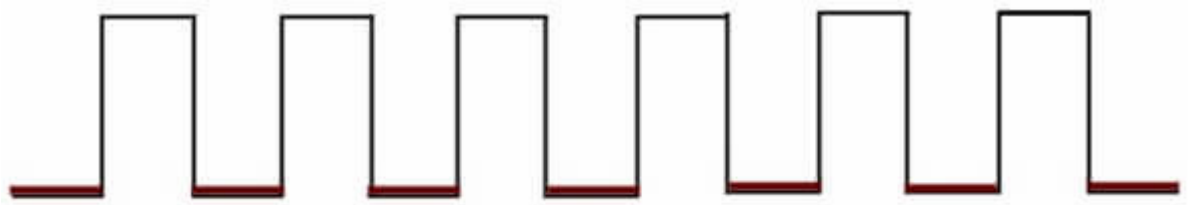
Positive level triggering

In a positive level triggering, the signal with Logic High occurs. So, in this triggering, the circuit is operated with such type of clock signal. Below is the diagram of positive level triggering:



Negative level triggering

In negative level triggering, the signal with Logic Low occurs. So, in this triggering, the circuit is operated with such type of clock signal. Below is the diagram of Negative level triggering:



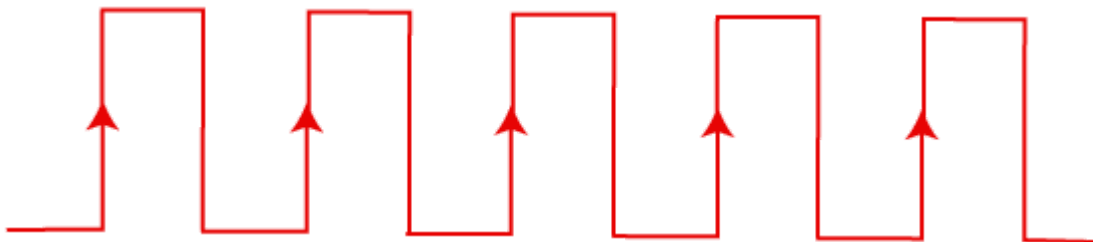
Edge triggering

In clock signal of edge triggering, two types of transitions occur, i.e., transition either from Logic Low to Logic High or Logic High to Logic Low.

Based on the transitions of the clock signal, there are the following types of edge triggering:

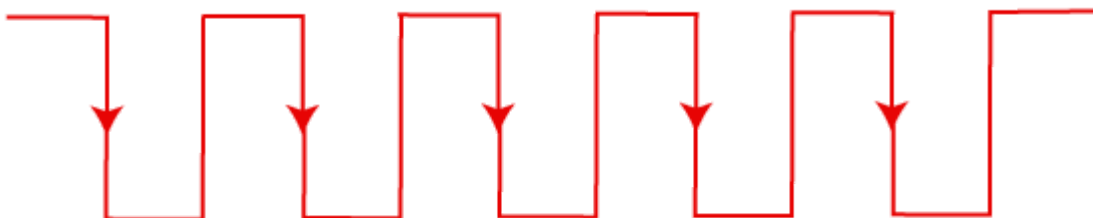
Positive edge triggering

The transition from Logic Low to Logic High occurs in the clock signal of positive edge triggering. So, in positive edge triggering, the circuit is operated with such type of clock signal. The diagram of positive edge triggering is given below.



Negative edge triggering

The transition from Logic High to Logic low occurs in the clock signal of negative edge triggering. So, in negative edge triggering, the circuit is operated with such type of clock signal. The diagram of negative edge triggering is given below.



Latches

Latches are **basic storage elements** that operate with signal levels (rather than signal transitions). **Latches controlled by a clock transition are flip-flops**. Latches are **level-sensitive devices**. Latches are useful for the design of

the asynchronous sequential circuit. Latches are sequential circuit with two stable states. These are sensitive to the input voltage applied and does not depend on the clock pulse. Flip flops that do not use clock pulse are referred to as latch.

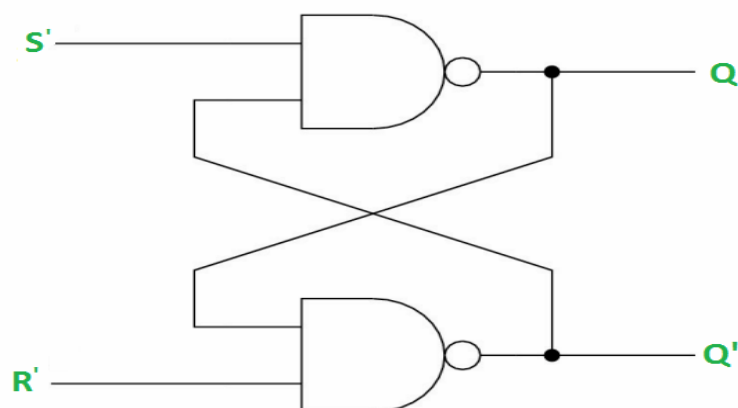
SR (Set-Reset) Latch – They are also known as **preset and clear states**. The SR latch forms the **basic building blocks of all other types of flip-flops**.

SR Latch is a circuit with:

- (i) 2 cross-coupled NOR gate or 2 cross-coupled NAND gate.
- (ii) 2 input S for SET and R for RESET.
- (iii) 2 output Q, Q'.

Q	Q'	STATE
1	0	Set
0	1	Reset

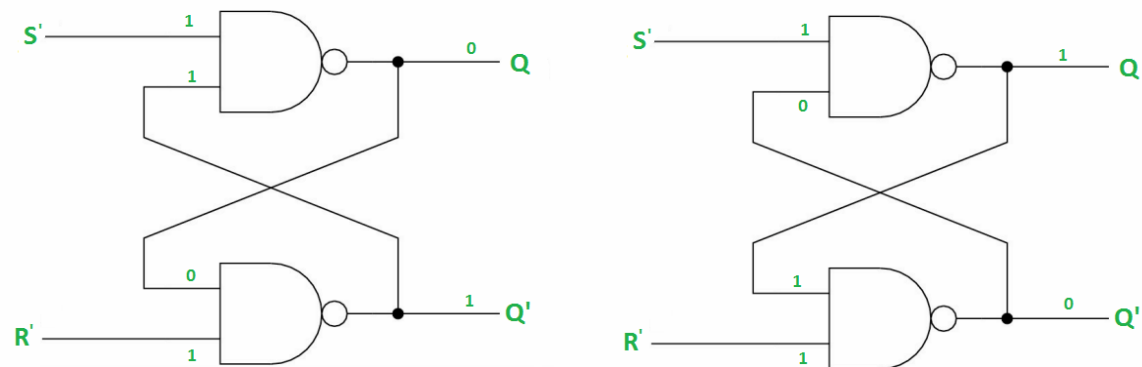
Under normal conditions, both the input remains 0. The following is the RS Latch with NAND gates:



Case-1: $S'=R'=1$ ($S=R=0$) –

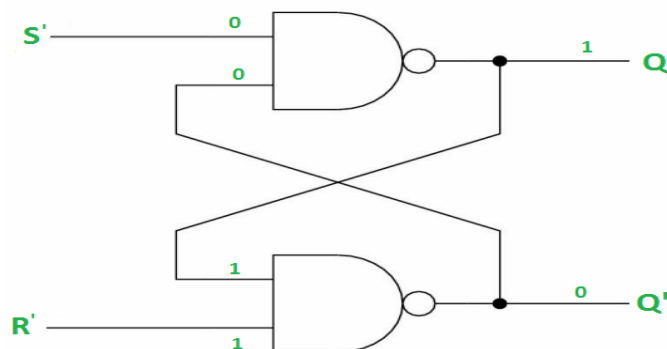
If $Q = 1$, Q and R' inputs for 2nd NAND gate are both 1.

If $Q = 0$, Q and R' inputs for 2nd NAND gate are 0 and 1 respectively.



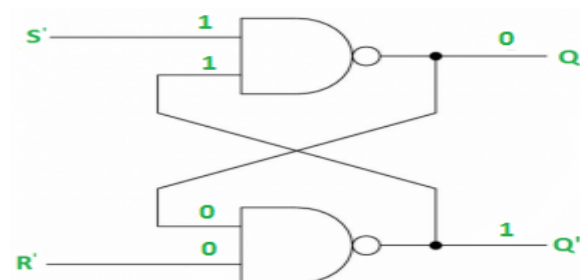
Case-2: $S'=0$, $R'=1$ ($S=1$, $R=0$) –

As $S'=0$, the output of 1st NAND gate, $Q = 1$ (**SET state**). In 2nd NAND gate, as Q and R' inputs are 1, $Q'=0$.



Case-3: $S'=1$, $R'=0$ ($S=0$, $R=1$) –

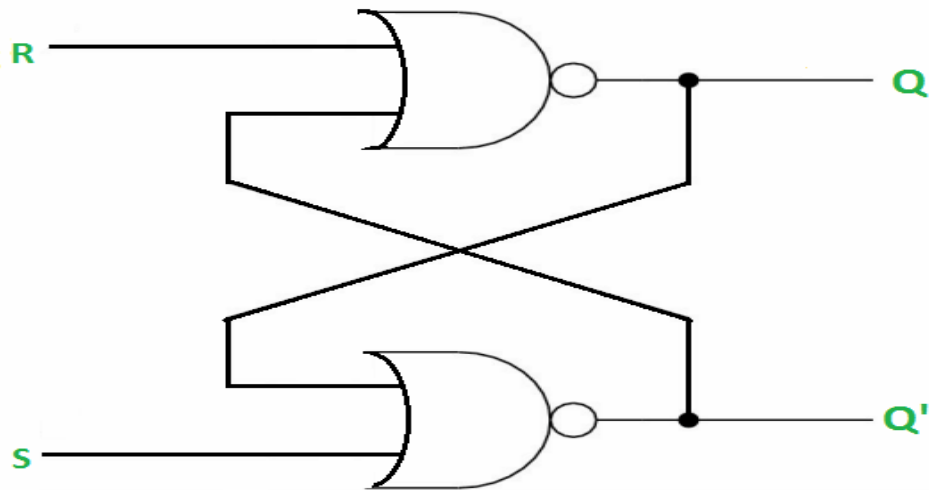
As $R'=0$, the output of 2nd NAND gate, $Q' = 1$. In 1st NAND gate, as Q and S' inputs are 1, $Q=0$ (**RESET state**).



Case-4: $S'=R'=0$ ($S=R=1$) –

When $S=R=1$, both Q and Q' becomes 1 which is not allowed. So, the input condition is prohibited.

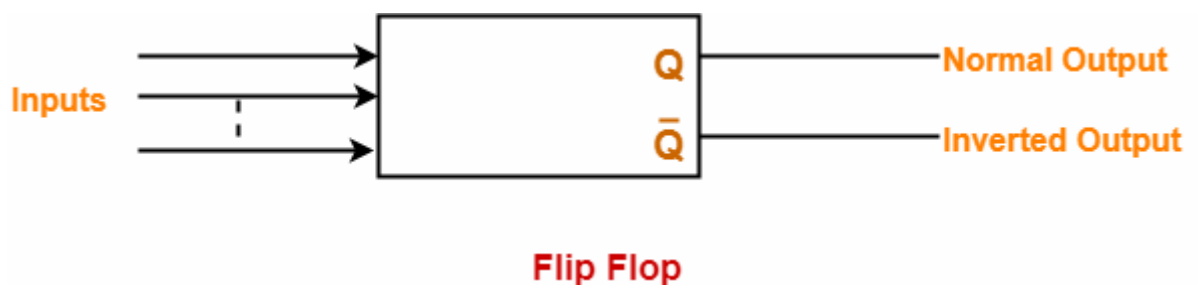
The SR Latch using NOR gate is shown below:



Flip Flop-

A Flip Flop is a memory element that is capable of storing one bit of information.

A flip flop has two outputs as shown-



A flip flop can maintain a binary state for an unlimited period of time as long as-

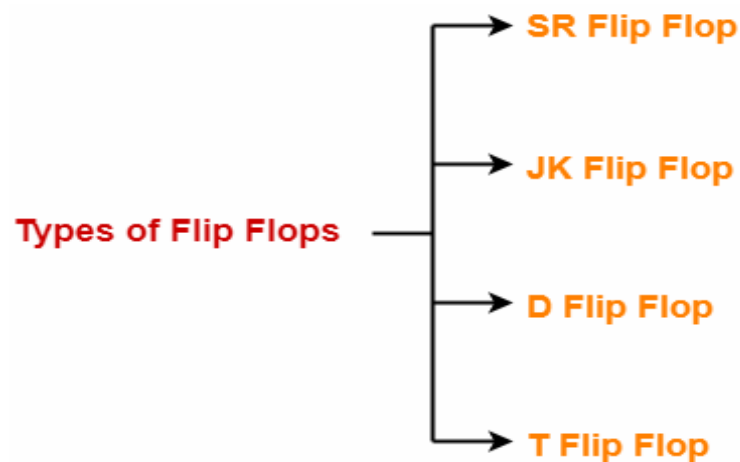
- Power is supplied to the circuit.
- Or until it is directed by an input signal to switch states.

A flip flop is also called as **Bistable Multivibrator** because it has two stable states either 0 or 1.

Flip Flops Types-

Flip flops are of different types depending on how their inputs and clock pulses cause transition between two states.

There are 4 basic types of flip flops-



1. SR Flip Flop
2. JK Flip Flop
3. D Flip Flop
4. T Flip Flop

SR Flip Flop-

- SR flip flop is the simplest type of flip flops.
- It stands for **Set Reset flip flop**.
- It is a clocked flip flop.

Construction of SR Flip Flop-

There are following two methods for constructing a SR flip flop-

Methods for Constructing SR Flip Flop

By Using NOR Latch

By Using NAND Latch

1. By using NOR latch
2. By using NAND latch

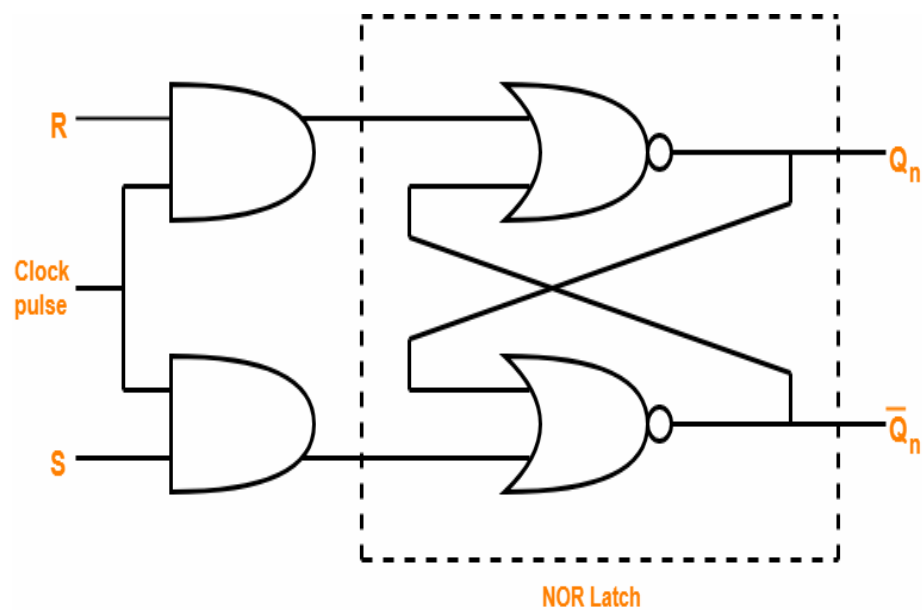
1. Construction of SR Flip Flop By Using NOR Latch-

This method of constructing SR Flip Flop uses-

- NOR latch
- Two AND gates

Logic Circuit-

The logic circuit for SR Flip Flop constructed using NOR latch is as shown below-



SR Flip Flop Using NOR Latch

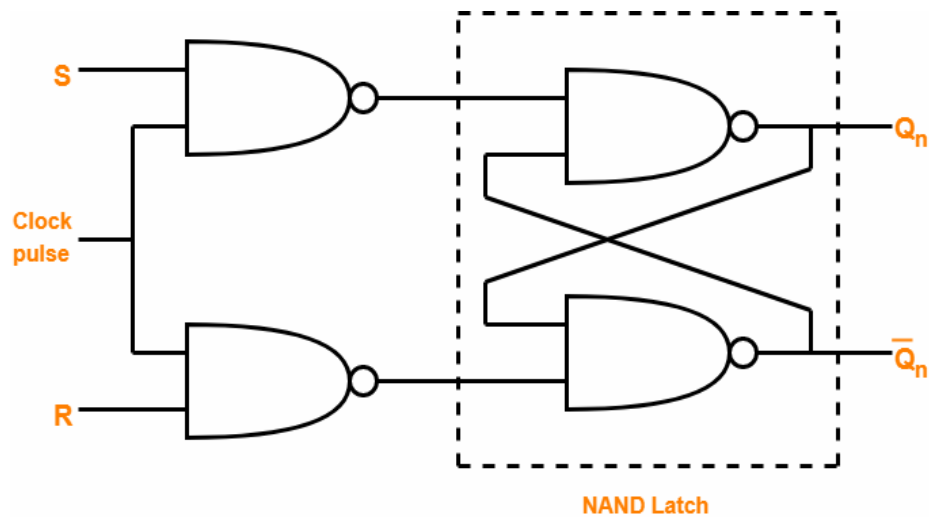
2. Construction of SR Flip Flop By Using NAND Latch-

This method of constructing SR Flip Flop uses-

- NAND latch
- Two NAND gates

Logic Circuit-

The logic circuit for SR Flip Flop constructed using NAND latch is as shown below-



SR Flip Flop Using NAND Latch

Logic Symbol-

The logic symbol for SR Flip Flop is as shown below-



Logic Symbol

Truth Table-

The truth table for SR Flip Flop is as shown below-

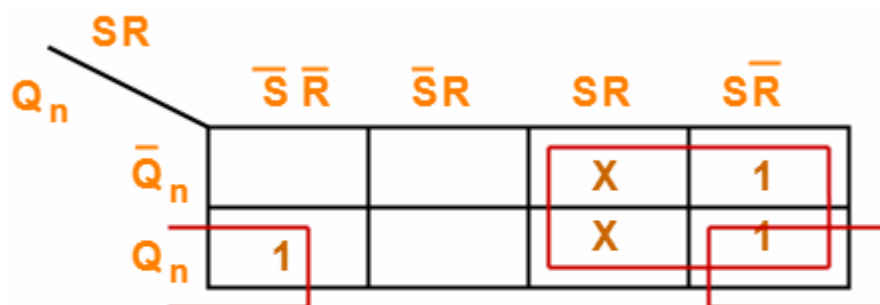
INPUTS			OUTPUTS
S	R	Q_n (Present State)	Q_{n+1} (Next State)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Indeterminate
1	1	1	Indeterminate

The above truth table may be reduced as-

INPUTS			OUTPUTS	REMARKS
S	R	Q_n (Present State)	Q_{n+1} (Next State)	States and Conditions
0	0	X	Q_n	Hold State condition $S = R = 0$
0	1	X	0	Reset state condition $S = 0$, $R = 1$
1	0	X	1	Set state condition $S = 1$, $R = 0$
1	1	X	Indeterminate	Indeterminate state condition $S = R = 1$

Characteristic Equation-

Draw a k map using the above truth table-



K Map

From here-

$$Q_{n+1} = (SR + SR') (Q_n + Q'_n) + Q_n (S'R' + SR')$$

$$Q_{n+1} = S + Q_n R'$$

Excitation Table-

The excitation table of any flip flop is drawn using its truth table.

What is excitation table?

For a given combination of present state Q_n and next state Q_{n+1} , excitation table tell the inputs required.

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

JK Flip Flop-

JK flip flop is a refined & improved version of **SR Flip Flop**

that has been introduced to solve the problem of indeterminate state that occurs in SR flip flop when both the inputs are 1.

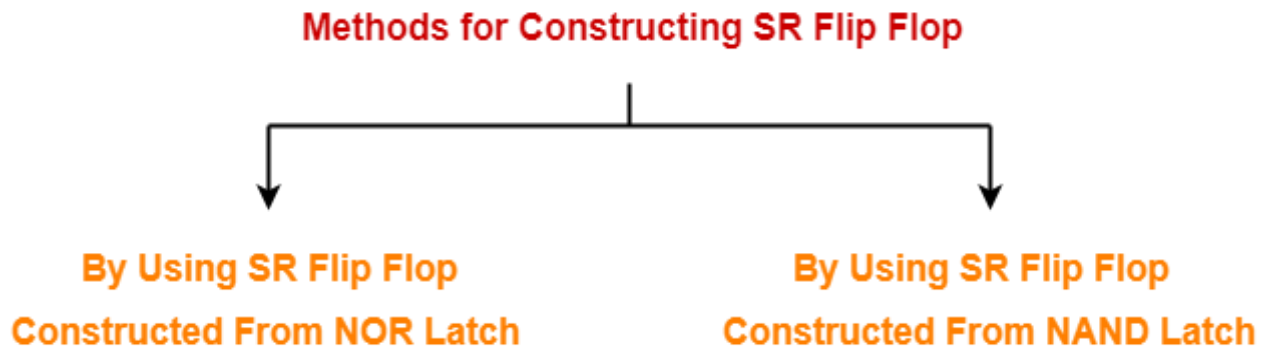
In JK flip flop,

- Input J behaves like input S of SR flip flop which was meant to set the flip flop.

- Input K behaves like input R of SR flip flop which was meant to reset the flip flop.

Construction of JK Flip Flop-

There are following two methods for constructing a JK flip flop-



1. By using SR flip flop constructed from NOR latch
2. By using SR flip flop constructed from NAND latch

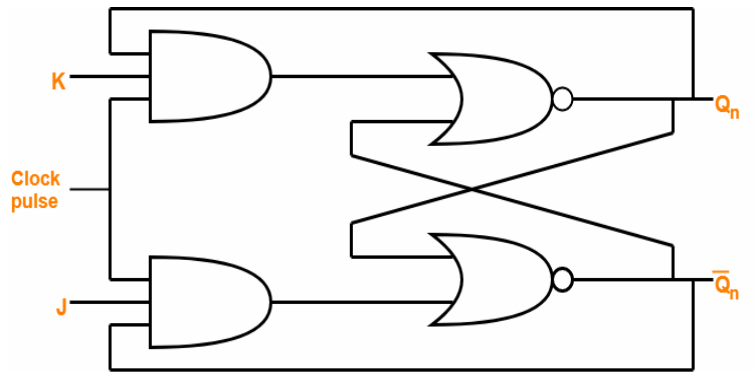
1. Construction of JK Flip Flop By Using SR Flip Flop Constructed From NOR Latch-

This method of constructing JK Flip Flop uses-

- SR Flip Flop constructed from NOR latch
- Two other connections

Logic Circuit-

The logic circuit for JK Flip Flop constructed using SR Flip Flop constructed from NOR latch is as shown below-



Logic Circuit For JK Flip Flop Using SR Flip Flop
(Constructed From NOR Latch)

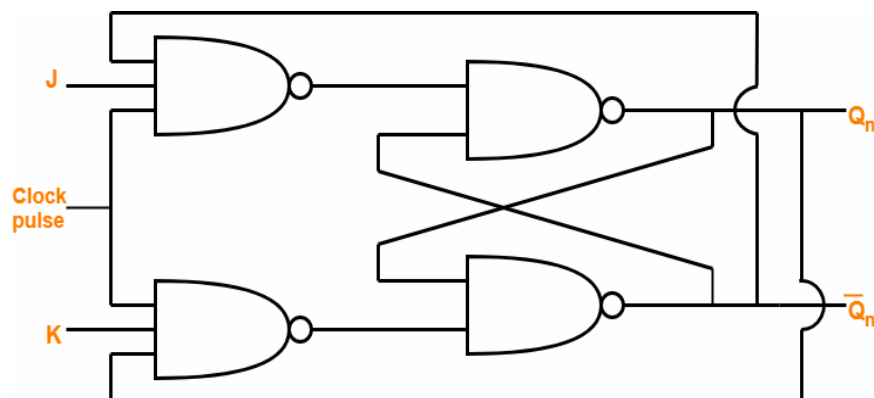
2. Construction of JK Flip Flop By Using SR Flip Flop Constructed From NAND Latch-

This method of constructing JK Flip Flop uses-

- SR Flip Flop constructed from NAND latch
- Two other connections

Logic Circuit-

The logic circuit for JK Flip Flop constructed using SR Flip Flop constructed from NAND latch is as shown below-



Logic Circuit For JK Flip Flop Using SR Flip Flop
(Constructed Using NAND Latch)

Logic Symbol-

The logic symbol for JK Flip Flop is as shown below-



Logic Symbol

Truth Table-

The truth table for JK Flip Flop is as shown below-

INPUTS			OUTPUTS
J	K	Q_n (Present State)	Q_{n+1} (Next State)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

The above truth table may be reduced as-

INPUTS			OUTPUTS	REMARKS
J	K	Q_n (Present State)	Q_{n+1} (Next State)	States and Conditions
0	0	X	Q_n	Hold State condition $J = K = 0$
0	1	X	0	Reset state condition $J = 0, K = 1$
1	0	X	1	Set state condition $J = 1, K = 0$
1	1	X	Q'_n	Toggle state condition $J = K = 1$

Characteristic Equation-

Draw a k map using the above truth table-



From here-

$$Q_{n+1} = Q'_n (JK + JK') + Q_n (J'K' + JK')$$

$$Q_{n+1} = Q'_n J + Q_n K'$$

Excitation Table-

The excitation table of any flip flop is drawn using its truth table.

What is excitation table?

For a given combination of present state Q_n and next state Q_{n+1} , excitation table tell the inputs required.

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Excitation Table

SR Flip Flop Vs JK Flip Flop-

Both JK flip flop and SR flip flop are functionally same.

The only difference between them is-

- In JK flip flop, indeterminate state does not occur.
- In JK flip flop, instead of indeterminate state, the present state toggles.

- In other words, the present state gets inverted when both the inputs are 1.

D Flip Flop

In **SR NAND Gate Bistable** circuit, the undefined input condition of SET = "0" and RESET = "0" is forbidden. It is the drawback of the SR flip flop. This state:

1. Override the feedback latching action.
2. Force both outputs to be 1.
3. Lose the control by the input, which first goes to 1, and the other input remains "0" by which the resulting state of the latch is controlled.

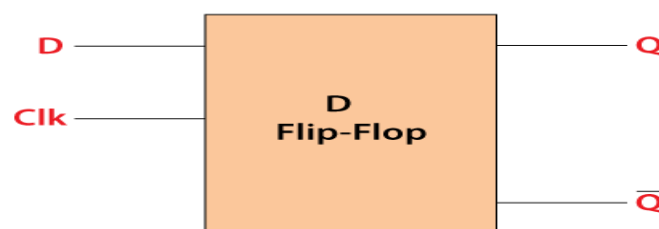
We need an **inverter** to prevent this from happening. We connect the inverter between the Set and Reset inputs for producing another type of flip flop circuit called **D flip flop**, Delay flip flop, D-type Bistable, D-type flip flop.

The D flip flop is the most important flip flop from other clocked types. It ensures that at the same time, both the inputs, i.e., S and R, are never equal to 1. The Delay flip-flop is designed using a gated SR flip-flop with an inverter connected between the inputs allowing for a single input D(Data).

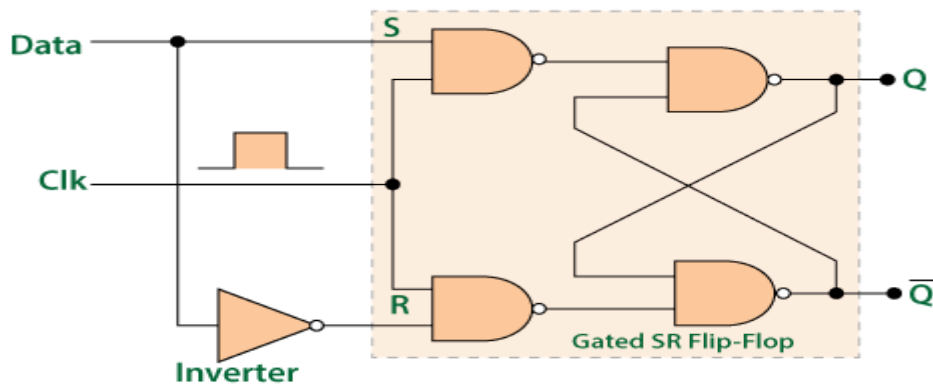
This single data input, which is labeled as "D" used in place of the "Set" input and for the complementary "Reset" input, the inverter is used. Thus, the level-sensitive D-type or D flip flop is constructed from a level-sensitive SR flip flop.

So, here $S=D$ and $R=\sim D$ (complement of D)

Block Diagram



Circuit Diagram



We know that the SR flip-flop requires two inputs, i.e., one to "SET" the output and another to "RESET" the output. By using an inverter, we can set and reset the outputs with only one input as now the two input signals complement each other. In SR flip flop, when both the inputs are 0, that state is no longer possible. It is an ambiguity that is removed by the complement in D-flip flop.

In D flip flop, the single input "D" is referred to as the "Data" input. When the data input is set to 1, the flip flop would be set, and when it is set to 0, the flip flop would change and become reset. However, this would be pointless since the output of the flip flop would always change on every pulse applied to this data input.

The "CLOCK" or "ENABLE" input is used to avoid this for isolating the data input from the flip flop's latching circuitry. When the clock input is set to true, the D input condition is only copied to the output Q. This forms the basis of another sequential device referred to as **D Flip Flop**.

When the clock input is set to 1, the "set" and "reset" inputs of the flip-flop are both set to 1. So it will not change the state and store the data present on its output before the clock transition occurred. In simple words, the output is "latched" at either 0 or 1.

Truth Table for the D-type Flip Flop

Clock	D	Q	Q'	Description
↓ » 0	X	Q	Q'	Memory no change
↑ » 1	0	0	1	Reset Q » 0
↑ » 1	1	1	0	Set Q » 1

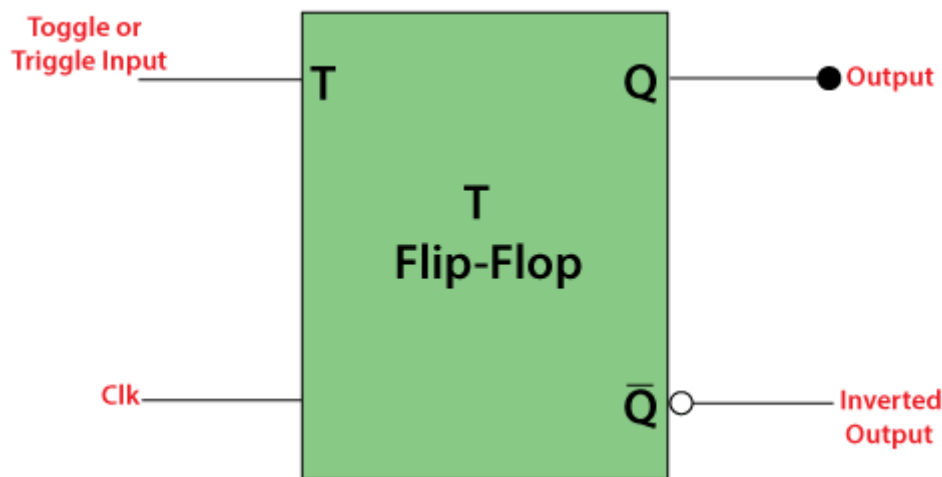
Symbols $\downarrow$ and $\uparrow$ indicates the direction of the clock pulse. D-type flip flop assumed these symbols as edge-triggers.

T Flip Flop

In T flip flop, "T" defines the term "Toggle". In SR Flip Flop, we provide only a single input called "Toggle" or "Trigger" input to avoid an intermediate state occurrence. Now, this flip-flop work as a Toggle switch. The next output state is changed with the complement of the present state output. This process is known as "Toggling".

We can construct the "T Flip Flop" by making changes in the "JK Flip Flop". The "T Flip Flop" has only one input, which is constructed by connecting the input of JK flip flop. This single input is called T. In simple words, we can construct the "T Flip Flop" by converting a "JK Flip Flop". Sometimes the "T Flip Flop" is referred to as single input "JK Flip Flop".

Block diagram of the "T-Flip Flop" is given where T defines the "Toggle input", and CLK defines the clock signal input.



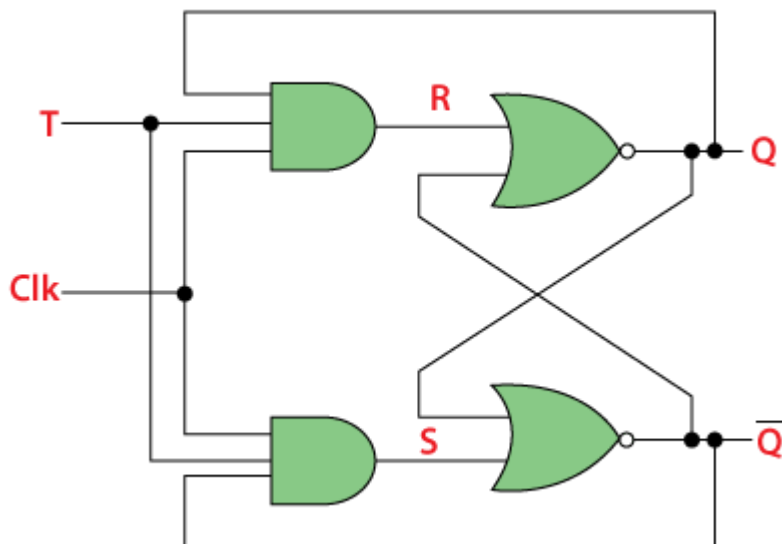
T Flip Flop Circuit

There are the following two methods which are used to form the "T Flip Flop":

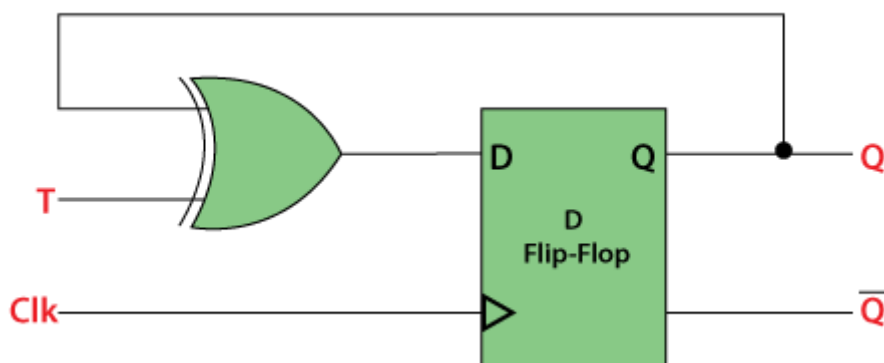
- By connecting the output feedback to the input in "SR Flips Flop".
- We pass the output that we get after performing the XOR operation of T and Q_{PREV} output as the D input in D Flip Flop.

Construction

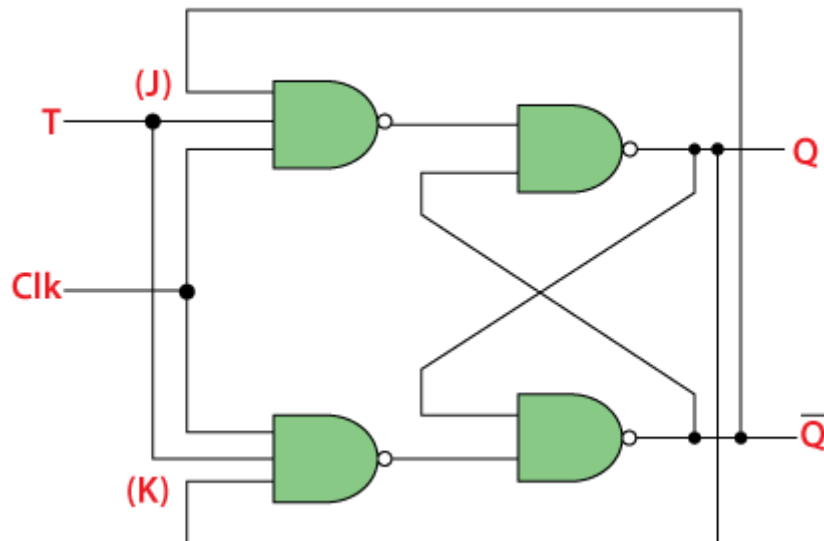
The "T Flip Flop" is designed by passing the AND gate's output as input to the NOR gate of the "SR Flip Flop". The inputs of the "AND" gates, the present output state Q , and its complement Q' are sent back to each AND gate. The toggle input is passed to the AND gates as input. These gates are connected to the Clock (CLK) signal. In the "T Flip Flop", a pulse train of narrow triggers are passed as the toggle input, which changes the flip flop's output state. The circuit diagram of the "T Flip Flop" using "SR Flip Flop" is given below:



The "T Flip Flop" is formed using the "D Flip Flop". In D flip - flop, the output after performing the [XOR](#) operation of the T input with the output " Q_{PREV} " is passed as the D input. The logical circuit of the "T-Flip Flop" using the "D Flip Flop" is given below:



The simplest construction of a T Flip Flop is with JK Flip Flop. Both the inputs of the "JK Flip Flop" are connected as a single input T . Below is the logical circuit of the T Flip Flop" which is formed from the "JK Flip Flop":



Truth Table of T Flip Flop

	Previous		Next	
T	Q	Q'	Q	Q'
0	0	1	0	1
0	1	0	1	0
1	0	1	1	0
1	1	0	0	1

The upper NAND gate is enabled, and the lower NAND gate is disabled when the output Q To is set to 0. make the flip flop in "set state(Q=1)", the trigger passes the S input in the flip flop.

The upper NAND gate is disabled, and the lower NAND gate is enabled when the output Q is set to 1. The trigger passes the R input in the flip flop to make the flip flop in the reset state(Q=0).

Operations of T-Flip Flop

The next sate of the T flip flop is similar to the current state when the T input is set to false or 0.

- If toggle input is set to 0 and the present state is also 0, the next state will be 0.
- If toggle input is set to 0 and the present state is 1, the next state will be 1.

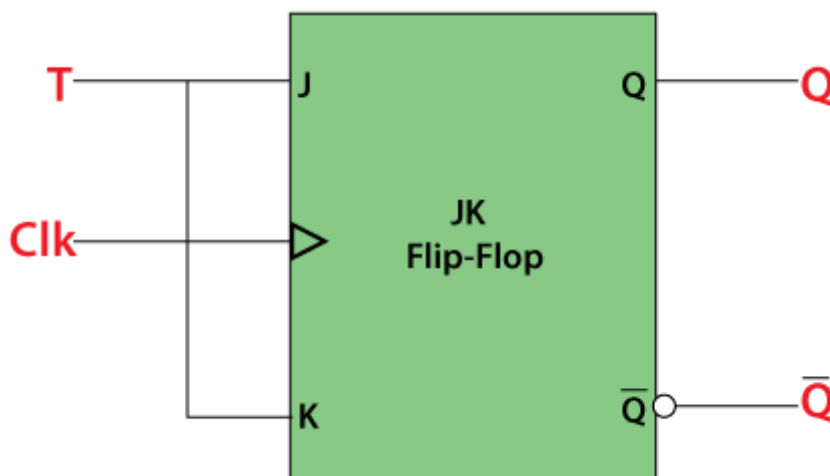
The next state of the flip flop is opposite to the current state when the toggle input is set to 1.

- If toggle input is set to 1 and the present state is 0, the next state will be 1.
- If toggle input is set to 1 and the present state is 1, the next state will be 0.

The "T Flip Flop" is toggled when the set and reset inputs alternatively changed by the incoming trigger. The "T Flip Flop" requires two triggers to complete a full cycle of the output waveform. The frequency of the output produced by the "T Flip Flop" is half of the input frequency. The "T Flip Flop" works as the "Frequency Divider Circuit."

In "T Flip Flop", the state at an applied trigger pulse is defined only when the previous state is defined. It is the main drawback of the "T Flip Flop".

The "T flip flop" can be designed from "JK Flip Flop", "SR Flip Flop", and "D Flip Flop" because the "T Flip Flop" is not available as ICs. The block diagram of "T Flip Flop" using "JK Flip Flop" is given below:



Master-Slave JK Flip Flop

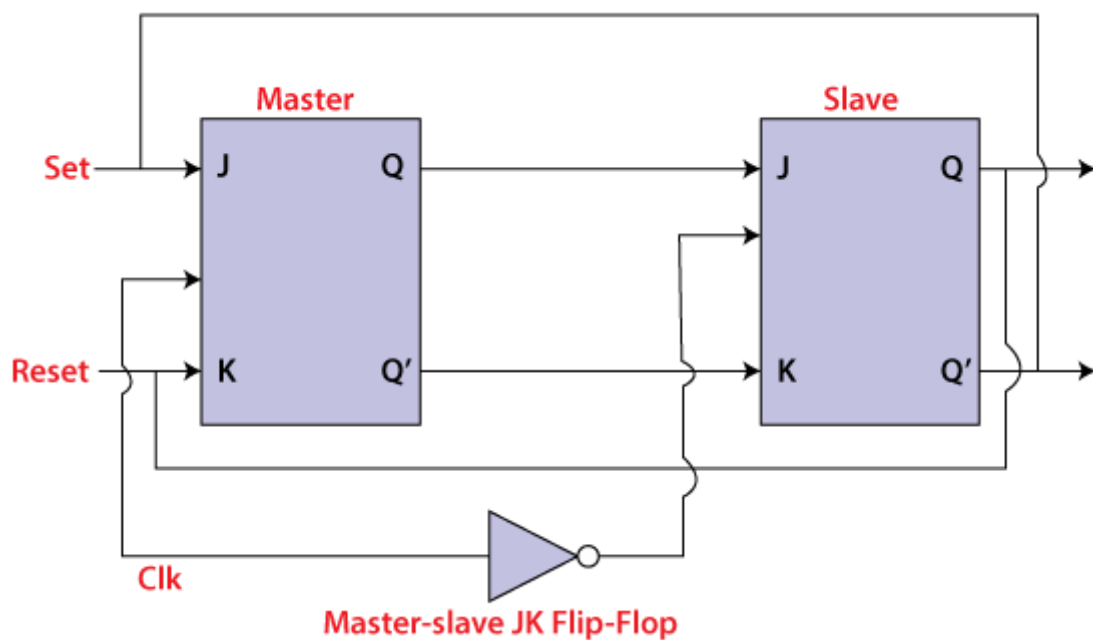
In "JK Flip Flop", when both the inputs and CLK set to 1 for a long time, then Q output toggle until the CLK is 1. Thus, the uncertain or unreliable output produces. This problem is referred to as a race-round condition in JK flip-flop and avoided by ensuring that the CLK set to 1 only for a very short time.

Explanation

The master-slave flip flop is constructed by combining two JK flip flops. These flip flops are connected in a series configuration. In these two flip flops, the 1st flip flop work as "master", called the master flip flop, and the 2nd work as a

"slave", called slave flip flop. The master-slave flip flop is designed in such a way that the output of the "master" flip flop is passed to both the inputs of the "slave" flip flop. The output of the "slave" flip flop is passed to inputs of the master flip flop.

In "master-slave flip flop", apart from these two flip flops, an inverter or NOT gate is also used. For passing the inverted clock pulse to the "slave" flip flop, the inverter is connected to the clock's pulse. In simple words, when CP set to false for "master", then CP is set to true for "slave", and when CP set to true for "master", then CP is set to false for "slave".



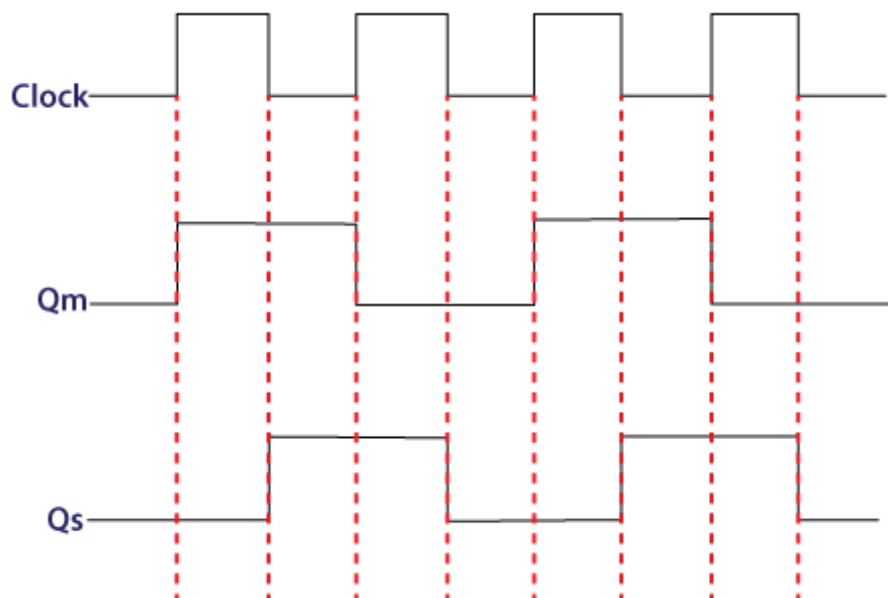
Working:

- When the clock pulse is true, the slave flip flop will be in the isolated state, and the system's state may be affected by the J and K inputs. The "slave" remains isolated until the CP is 1. When the CP set to 0, the master flip-flop passes the information to the slave flip flop to obtain the output.
- The master flip flop responds first from the slave because the master flip flop is the positive level trigger, and the slave flip flop is the negative level trigger.
- The output $Q'=1$ of the master flip flop is passed to the slave flip flop as an input K when the input J set to 0 and K set to 1. The clock forces the

slave flip flop to work as reset, and then the slave copies the master flip flop.

- When $J=1$, and $K=0$, the output $Q=1$ is passed to the J input of the slave. The clock's negative transition sets the slave and copies the master.
- The master flip flop toggles on the clock's positive transition when the inputs J and K set to 1. At that time, the slave flip flop toggles on the clock's negative transition.
- The flip flop will be disabled, and Q remains unchanged when both the inputs of the JK flip flop set to 0.

Timing Diagram of a Master Flip Flop:



- When the clock pulse set to 1, the output of the master flip flop will be one until the clock input remains 0.
- When the clock pulse becomes high again, then the master's output is 0, which will be set to 1 when the clock becomes one again.
- The master flip flop is operational when the clock pulse is 1. The slave's output remains 0 until the clock is not set to 0 because the slave flip flop is not operational.
- The slave flip flop is operational when the clock pulse is 0. The output of the master remains one until the clock is not set to 0 again.

- Toggling occurs during the entire process because the output changes once in the cycle.

Counter in Digital Electronics

A counter is a digital circuit that can be used to keep track of the number of events or clock cycles that have occurred. Counters can be classified into different types based on their counting sequence and the number of bits they can count. Some common types of counters include binary counters, decade counters, and up/down counters.

A binary counter is a type of counter that counts in binary, and the number of bits it uses determines its maximum count. For example, a 4-bit binary counter can count up to 15 (in decimal), and an 8-bit binary counter can count up to 255 (in decimal).

A decade counter is a type of counter that counts in decimal, and it has 10 states. Decade counters are often used in digital displays, where they can be used to drive the digits of a 7-segment display.

An up/down counter is a type of counter that can count up or down, depending on the control signals. Up/down counters are often used in digital systems where both counting up and down are required.

In digital electronics, counters are often implemented using flip-flops, and they play a crucial role in many digital systems, such as timers, digital clocks, and frequency counters.

Register in Digital Electronics

A register in digital electronics is a group of flip-flops that can be used to store binary data. Registers are often used to hold intermediate results in digital systems, or to store the state of a system for a period of time.

The number of flip-flops in a register determines the size of the register, and the size of the register determines the number of bits that can be stored in the register. For example, a 4-bit register can store a 4-bit binary number, while an 8-bit register can store an 8-bit binary number.

Registers can be used in many different applications, including memory systems, arithmetic logic units (ALUs), and input/output (I/O) interfaces. In

addition, registers can be used in combination with counters to form larger memory systems, such as RAM and ROM.

The operation of a register is typically controlled by a set of control signals, such as load, shift, and enable signals. The load signal is used to transfer data into the register, while the shift signal is used to shift the data within the register. The enable signal is used to control the operation of the register, and it can be used to enable or disable the register.